Clark Zhang

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Work Authorization: Canadian Citizen — Eligible for U.S. TN Status (No H-1B required)

EDUCATION

University of British Columbia **Graduation: April 2026**
Bachelor of Applied Science - Integrated Engineering (Mechatronics Specialization), Co-op Program
Cumulative GPA: 4.0/4.0 (92%)

SKILLS

- **Mechanical Design:** SolidWorks (CSWP), CATIA, Fusion 360, AutoCAD, SolidWorks FEA/Flow, Ansys FEA/Fluent, GD&T
- **Robotics & Controls:** PLC Control (UR/KUKA), PCB design (Altium), Python, C++, ROS2, Git, MATLAB, RoboDK
- **Manufacturing & Prototyping:** CNC, waterjet/laser cutting, 3D printing, composites, shop tools, shop-ready drawings

WORK EXPERIENCE

Engineering Project Lead – Mechatronics, Hein Lab **December 2024 – January 2026**
- Directed a 6-person engineering team developing a robotic drug-analysis platform for Canada's overdose crisis response, automating sample preparation and downstream analysis of 1500+ samples with 98% overall reliability
- Designed, fabricated, and tested robotic end effectors and automation modules in SolidWorks/Fusion 360, including a vial gripper, vortexer, decapper, liquid handler, and BLDC centrifuge; reduced module costs from $1,500-$5,000 to <$100
- Produced 11 CAD assemblies and GD&T manufacturing drawings for precision lab-grade components fabricated through 3D printing, CNC machining, and waterjet cutting
- Designed and fabricated 2 custom PCBs in Altium for synchronized stepper/BLDC motor control and sensor integration
- Developed a multi-tiered control architecture using PLC/Python to orchestrate UR robotic arm motion and high-level workflows, integrated with C++/ROS2 for hardware module automation and sensor error detection

Mechatronics Engineering Intern, Hein Lab **April 2024 – December 2024**
- Commissioned 4 UR/KUKA industrial robot arms across 3 deployed automation platforms, handling mounting, I/O wiring, networking, reach validation, collision checks, and end-to-end debugging while reducing operator hands-on time by ~80%
- Transitioned hard-to-maintain PLC-based robot sequences into reusable Python control libraries for UR and KUKA arms, making robotic workflows faster to modify, test, and deploy
- Modeled robotic workcells in SolidWorks and simulated workflows in RoboDK to verify reachability, collision clearance, and module access before deployment, reducing workflow errors by 5x

Airfoils Team Lead, UBC AeroDesign **April 2022 – April 2026**
- Led an 11-person subteam designing and manufacturing a 120-inch heavy-lift RC aircraft for the SAE AeroDesign competition, overseeing wing design, assembly integration, manufacturing, and testing; earned 2nd place overall in 2025
- Designed main wing, ailerons, servo mounts, and linkage interfaces in SolidWorks and CATIA V5 using DFMA and GD&T principles, allowing clean integration into overall assembly
- Validated aerodynamic and structural performance using SolidWorks FEA/Flow, ANSYS, and wind-tunnel testing, improving lift performance by ~20% before manufacturing
- Manufactured flight hardware using 3D printing, CNC machining, waterjet/laser cutting, and carbon-fiber prepreg layups, standardizing assembly processes and reducing build errors by 70%

PROJECTS

Robotic Tool Vending Machine (Haven) **September 2025 – April 2026**
- Designed and validated a Cartesian XY gantry and rotary indexing storage mechanism in SolidWorks FEA, including lead-screw motion, bearing interfaces, structural supports, and manufacturable sheet/3D-printed components
- Fabricated and iterated prototypes using 3D printing, waterjet-cut plates, CNC-machined parts, hand tools, and fit testing
- Developed Raspberry Pi/Klipper motion control with magnetic encoder feedback and a custom I2C multiplexer PCB to coordinate gantry travel, storage indexing, and tool-position verification

Semi-Autonomous Aquatic Trash-Removal Robot **September 2024 – April 2025**
- Designed and fabricated aquatic mechanical assemblies in SolidWorks, including a dual-pontoon flotation chassis, structural supports, and adjustable propulsion mounts; won 1st place at the National CSME Design Competition
- Integrated sealed plywood, aluminum, and 3D-printed parts into a watertight prototype optimized for saltwater operation